ACS NANO
Cite This: ACS Nano XXXX, XXX, XXX–XXX
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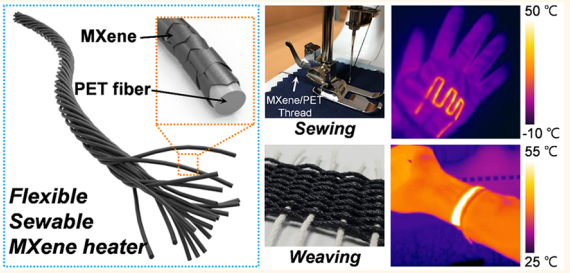
Shape-Adaptable 2D Titanium Carbide (MXene) Heater

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Supporting Information

ABSTRACT: **Prior to the advent of the next-generation heater for wearable/on-body electronic devices, various properties are required, including conductivity, transparency, mechanical reliability, and conformability. Expansion to two-dimensional (2D) structure of metallic nanowires based on network- and mesh-type geometries has been widely exploited for realizing these heaters. However, the routes led to many drawbacks such as the low-density cross-bar linking, self-aggregation of wire, and high junction resistance. Although 2D carbon nanomaterials such as graphene and reduced graphene oxide (rGO) have shown their potentials for the purpose, CVD-grown graphene with sufficiently high conductivity was limited due to its poor processability for large-area applications, while rGO fabricated with a complex reduction process involving the use of toxic chemicals suffered from a low electrical conductivity. In this study, we demonstrate a simple and robust process, utilizing electrostatic assembling of negatively charged MXene flakes on a positively treated surface of substrate, for fabricating a metal-like 2D MXene thin film heater (TFH). Our TFH showed a high optical property (>65%), low sheet resistance (215 Ω/sq), fast electrothermal response (within dozens of seconds) with an intrinsically high electrical conductivity, and mechanical flexibility (up to 180° bending). Its capability for forming a firm and stable ionic-type interface with a counterpart surface allows us to develop a shape-adaptable and patchable thread heater (TH) that can be shaped on diverse substrates even under harsh conditions of conventional sewing or weaving processes. This work suggests that our shape-adaptable MXene heaters are potentially suitable not only for wearable devices for local heating and defrosting but also for a variety of emerging applications of soft actuators and wearable/flexible healthcare monitoring and thermotherapy.**



KEYWORDS: *solution-processed MXene, two-dimensional nanomaterials, thin-film heater, shape-adaptable heater, thread heater, sewable fiber heater*

Electric heaters based on the conversion of electrical current into heat energy have been widely used for over a century in numerous personal and industrial applications, such as local heating,[1] automotive defrosting,[2,3] soft actuators,[4] drug release,[5] and micropatterning.[6] Recently, to respond to the demands of newly emerging heater applications, in particular, wearable/on-body electronic devices for healthcare sensing and monitoring, future heater materials require characteristics such as transparency, mechanical reliability under deformation, and facile adaptability on topologically structured surfaces.[5,7−9] Although conventional metals are efficient and commercially available and have a high electrical conductivity (~10^7 S/m), their large weight (>7 g/cm^3), optical

Received: February 27, 2019
Accepted: May 22, 2019

Downloaded by 219.217.246.127 at 18:56:04:982 on May 31, 2019
from https://pubs.acs.org/doi/10.1021/acsnano.9b01602.

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DOI: 10.1021/acsnano.9b01602
ACS Nano XXXX, XXX, XXX–XXX

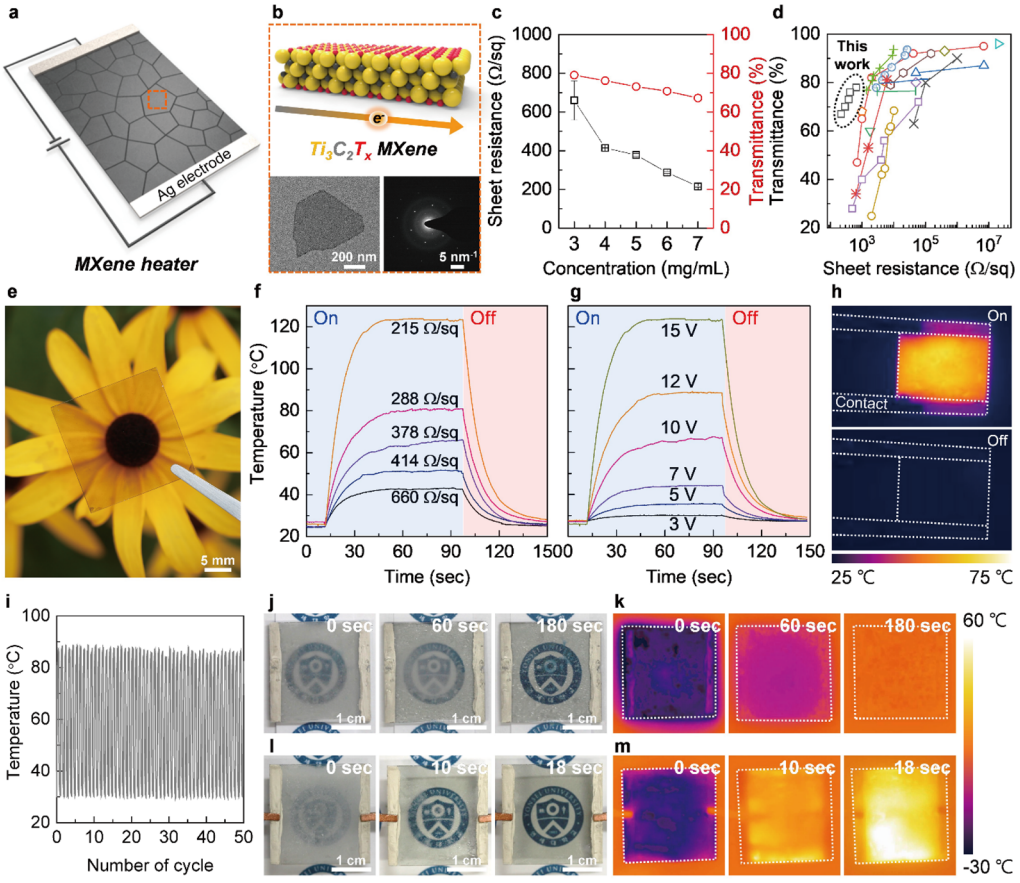


Figure 1. (a) Schematic of the TFH composed of MXene flakes. (b) Illustration of $Ti_3C_2T_x$ MXene, in which electrons move along its in-plane direction (top). TEM image (bottom left) and SAED pattern (bottom right) of the $Ti_3C_2T_x$ MXene flakes. (c) Plots of the sheet resistance and transmittance of the MXene TFH as a function of the MXene concentration in the solution. (d) Comparison plots of the sheet resistance and transmittance for the MXene TFH compared with those for the thin films based on rGO in the literature. Additional details are presented in Table S1. (e) Photograph of the MXene TFH on glass prepared using an MXene solution of 6 mg/mL. (f) Temperature variations of the MXene TFHs with different sheet resistance values as a function of time at 15 V. (g) Temperature variations of the MXene TFH prepared using the MXene solution of 6 mg/mL operated at different voltages as a function of time. (h) IR images of the MXene TFH in the ON (at 10 V, top) and OFF (bottom) states. (i) Temperature plots of the MXene TFH at 12 V under 50 repeated ON−OFF cycles. Photographs of the MXene TFH with a frosted surface without (j) and with (l) heating at 12 V, as well as the corresponding IR images (k and m, respectively). The symbol of Yonsei University was printed with permission from Yonsei University.

opaqueness, mechanical stiffness, and surface oxidation limit their applications.[10]

Heaters based on two dimensionally networked and/or meshed Ag nanowires resolved some of the issues aforementioned such as transparency as well as mechanical flexibility, but it is not still trivial to control the numerous cross-bar junctions arising from physical overlapping of one-dimensional nanowires required for high electrical conduction, which additionally requires cost-ineffective lithography processes with limited scales.[11] Moreover, for decades, low-dimensional carbon nanomaterials such as carbon nanotubes (CNTs)[10−18] and graphene[19−24] have received attention for electric heaters owing to their high electrical conductivity even with a low density (<2.2 g/cm^3). However, despite the excellent electrical and optical properties of these materials, they are restricted by their poor processability. As an alternative, reduced graphene oxide (rGO)[25−30] has been introduced, which has many potential advantages due to the surface oxi-functional groups allowing simple hybridization or coating processes on other materials owing to the high dispersion in solution. However, the decreased electrical conductivity due to the destruction of the atomic arrangement during oxidation has been an obstacle for electronic devices that require a high conductivity.[31] Moreover, the reduction processes of rGO are complex and involve the use of harmful and toxic chemicals.[32]

Herein, we report multifunctional heaters using the 2D transition-metal carbide, which is called MXene. $Ti_3C_2T_x$ MXene, which was first introduced by Prof. Gogotsi and Barsoum's group,[33] exhibits not only metal-like conductivity of approximately 10^6 S/m but also thickness-dependent electric-to-heat conversion behaviors of Ti_3C_2 MXene flakes validated from thermal microscopy analysis.[34] Additionally, the existence of a large number of terminal groups (−OH) forming the hydrophilic surface allows a simple solution process in water.[35] Therefore, an MXene-based electric heater could be an alternative to the previous 2D-structured heaters. In this study, we developed a large-area MXene thin-film heater (TFH) on a rigid glass substrate, which showed a fast electrothermal response time, high transparency, and high scalability. Then, we prepared the MXene-based heater mounted on a flexible poly(ethylene terephthalate) (PET) substrate, which exhibited both physical stability and excellent heating performance under mechanical deformation. For practical applications, we simply coated the thin MXene flakes

DOI: 10.1021/acsnano.9b01602
ACS Nano XXXX, XXX, XXX−XXX

on the surface of PET fibers treated via amino-silanization. MXene thread-based adaptable heaters with diverse shapes were demonstrated through sewing or weaving processes, in which the MXene flakes were strongly adhered to the PET fibers. When it was conformally attached to human skin, the flexible and mechanically adaptable MXene thread heater (TH) efficiently increased the body temperature.

RESULTS AND DISCUSSION

Transparent MXene Thin Film Heater. The MXene TFH was simply fabricated by one-step spin-coating of MXene-dispersed water after the hydrophilic silanization of a glass substrate for assembly with MXene flakes. The detailed procedures for the MXene synthesis and the preparation of the MXene TFH on the glass substrate are presented in the Experimental Section. The structure of our MXene TFH composed of uniformly distributed MXene flakes on glass substrates between two in-plane Ag electrodes is illustrated in Figure 1a. A single MXene flake is made up of alternating Ti and C elemental layers retaining terminal groups (T_x, T_x = −OH, −O, −F) on their surfaces, as shown in Figure 1b. The crystallographic characteristics of the MXene flakes were successfully identified using high-resolution transmission electron microscopy (HR-TEM) and its selected-area electron diffraction (SAED), which revealed that the MXene had a 2D shape and a single-crystalline nature. Figure 1c shows the electrical and optical properties of the MXene thin film with respect to the concentration of MXene in the solution (Figure S1). The sheet resistance values of the MXene thin film decreased with the increase of the concentration of MXene owing to the increased electrical conductivity arising from the electron density of states close to the Fermi level.[36] Moreover, although the optical transmittance at a wavelength of 550 nm decreased linearly with the increase of the concentration of MXene, it remained above 60% even at a sheet resistance of 200 Ω/sq, which could be sufficient for various optoelectronic applications. It should be noted that our MXene TFH exhibited lower sheet resistance values than TFHs comprising rGO and rGO/graphite hybrids, as shown in Figure 1d (Table S1). Figure 1e shows the transparency of our MXene TFH with an MXene concentration of 6 mg/mL.

The heating ability of the MXene TFH was evaluated by measuring the temperature changes according to time. At 15 V, the maximum temperature (T_{max}) value gradually increased with the decrease of the sheet resistance, which was determined by the concentration of MXene, as shown in Figure 1f. The MXene TFH with a sheet resistance value of 215 Ω/sq exhibited the highest T_{max} of approximately 120 °C, with an incremental average slope of 8.18 °C/s and recovery slope of −7.44 °C/s, as shown in Figure 1g. The average recovery rate values (Figure S2) were calculated from the slopes at a certain time right after the time reaching the maximum temperatures of the time−temperature plots in Figure 1g. The faster recovery rate was obtained due to the rapid heat dissipation at the high temperature. Since the temperature of the heater was proportional to the applied voltage (Figure 1g), the recovery rates increased with the applied voltage, consistent with the previous works.[7,16,24] The steady-state temperature was governed by the Joule heating of the electrothermal layer, which depended on the sheet resistance. The heat generation was proportional to the inverse of the sheet resistance, according to Joule's law: $H = V^2t/R$, where V is the input voltage, t is time, and R is the sheet resistance (Figure S3). The heating performance of the MXene TFH, which was determined using input voltage changes, agrees with the results for the sheet resistance from the viewpoint of the energy-dependent heating. The high value of T_{max} and fast response time within dozens of seconds allow our TFH to be utilized for high-performance thermal devices. During the heating state, because of the highly homogeneous dispersion of the MXene flakes on the glass substrate, the heater exhibited a uniform color change with the increase of the temperature without heat loss along the contact metal, as indicated by the pseudocolor images obtained using an infrared (IR) camera in Figure 1h. As shown in Figure 1i, our MXene TFH maintained its T_{max} values during 50 cycles of heating and cooling at 12 V, indicating its high durability at ambient condition. Moreover, after 50 cycles, the average resistance values of the MXene TFHs were rarely altered within the experimental uncertainty. The maximum temperatures reached at the corresponding applied voltages were hardly changed after 50 heating and cooling cycles. The results clearly show that our MXene TFHs were reproducible and reliable in their performance (Figure S4).

The thin-film high-resolution X-ray diffractometer (HR-XRD) patterns of MXene TFH before and after 50 cycles were examined with the heating temperature of approximately 90 °C in order to investigate alternation of the crystalline structure over a large area. The one dimensionally layered crystalline structure of the MXene TFH was still maintained even after repetitive heating and cooling cycles (Figure S5). The HR-XRD diffraction pattern of the MXene TFH before heating exhibited a strong peak at 6.3° corresponding to the (002) reflection of the basal plane of Ti_3C_2 with a c-lattice parameter of approximately 25 Å in its hexagonal lattice.[37] In addition, the high-order (00l) reflections apparent in the HR-XRD pattern suggest the long-range crystalline ordering of MXene flakes, which resulted from the compact stacking of the layers without the interruption of any intercalants. After the 50 cycles of heating and cooling, the crystalline structure was well maintained, but the intensities of the (00l) peaks including the (002) one were decreased. We believe that the long time exposure at the ambient conditions with the humidity during heating operation slightly reduced the crystallinity of the MXene TFH. It should be noted, however, that no secondary phases that frequently arose from the high-temperature oxidation of the MXene were observed during the multiple operation of the MXene TFH, giving rise to a consistent sheet resistance and heating performance (Figure S4).

As expected from the HR-XRD results, the X-ray photoelectron spectroscopy (XPS) spectrum of the MXene TFH after 50 cycles was rarely different from that before the cycles (Figure S6). The O 1s spectra of the MXene TFH showed the intense peaks located at approximately 529.6, 530.3, and 532.1 eV, corresponding to Ti−O, C−OH, and C−Ti−$(OH)_x$, respectively.[38] After 50 heating and cooling cycles, the peak intensity of the Ti−O bond was rarely altered, while that of C−Ti−$(OH)_x$ was slightly increased. In addition, the Ti 2p spectra of the MXene TFH revealed the presence of various Ti-related peaks of Ti−C, Ti(II), and Ti(III) and the weak peak of Ti−O (Figure S6c). Consistent with the results obtained from the O 1s spectra, there was no significant change in the peak of Ti−O. Meikang Han *et al.* reported that the secondary TiO_2 phases were frequently formed with thermal annealing of powders at least over 800 °C in an Ar environment.[38] Roghayyeh Lotfi *et al.* revealed that the MXene was durable even at 1000 K under damp air (relative humidity ~30%) within 30 s, while the rutile and anatase phases were evidently observed above 1500 K.[37]

DOI: 10.1021/acsnano.9b01602
ACS Nano XXXX, XXX, XXX−XXX

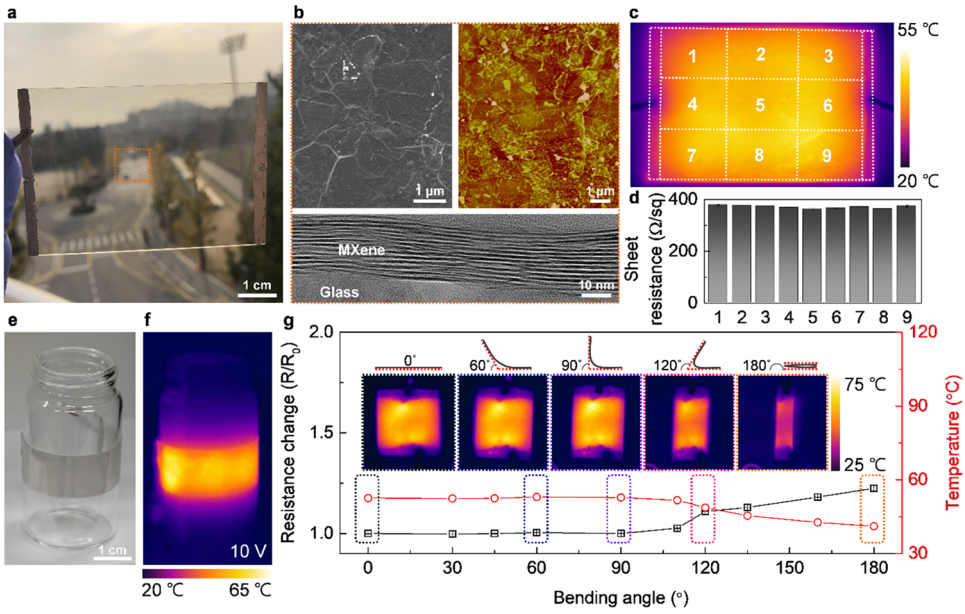


Figure 2. (a) Photograph of the large-area MXene TFH with a size of 5 cm × 7.5 cm. (b) Planar-view SEM (top left) and AFM (top right) images and cross-sectional TEM image (bottom) of the solution-processed, large-area MXene TFH prepared with a 5 mg/mL MXene solution. (c) IR image of the large-area MXene TFH at 25 V. (d) Distributions of the sheet resistance in the divided areas of the large-area MXene TFH, for the areas labeled 1 to 9 in (c). Photograph (e) and IR image (f) of the flexible MXene TFH prepared on the PET covering along the round surface of the glass vial. (g) Resistance and temperature change of the flexible MXene TFH as a function of the bending angle ranging from 0° to 180°. The inset shows IR images and schematics of the MXene TFH during bending.

The oxidation was accelerated in a humid environment with temperature. Importantly, Zhengyang Li *et al.* showed that the edge-site of MXene flakes could be slightly oxidized after heating at 200 °C under an O_2 atmosphere.[39] Considering that our MXene TFH was operated at its maximum temperature of 90 °C, the secondary TiO_2 phases were hardly formed on the basal plane.

The Raman spectrum of the MXene TFH before heating showed characteristic peaks at approximately 198, 402, and 602 cm^{-1}, which corresponded to vibration of carbon and oxygen bonds and Ti−C vibration of MXene (Figure S7).[40,41] After 50 heating and cooling cycles, the peaks were little changed. Furthermore, the characteristic peaks related to vibrational modes of anatase phases of crystalline TiO_2 were hardly observed at 144, 394, 513, and 635 cm^{-1}, corresponding to $E_{g(1)}$, $B_{1g(1)}$, A_{1g} and $B_{1g(2)}$, and $E_{g(3)}$, respectively.[42] HR-XRD, XPS, and Raman results showed that although slight chemical modification on the surface of our MXene TFH occurred, the secondary crystalline phases were rarely developed during repetitive heating and cooling cycles. Also, the MXene TFH was reliable after the cycles, reaching a maximum temperature of approximately 90 °C under ambient conditions.

Furthermore, we demonstrated the deicing performance by using our electric heater as a substitute for a conventional defroster, as shown in Figures 1j−m. This test was performed by inducing a voltage after immersing the MXene TFHs into liquid nitrogen for 5 min. Figure 1j and k show that without the voltage the frost on the surface was very slowly reduced for 180 s, and the actual temperature of the surface recovered from −30 °C to room temperature. On the other hand, at 12 V, the frost on the surface of the MXene TFH was rapidly removed, as shown in Figure 1l and m, recognizing clearly the background symbol without any residual water with the increase of the surface temperature to approximately 60 °C with an extremely fast response (Figure S8).

Large-Area and Flexible MXene TFH. MXene flakes of 5 mg/mL were uniformly distributed on large glass substrates (7.5 × 5 cm^2) and exhibited an optically high transparency over the whole area, as shown in Figure 2a. Scanning electron microscopy (SEM) and atomic force microscopy (AFM) images (Figure 2b, top) show uniformly distributed MXene flakes with a lateral size of approximately 1 μm over the entire glass substrate, without any voids. The root-mean-square surface roughness value for this MXene TFH was 2.45 nm, confirming its good flatness. Moreover, the MXene layers were investigated *via* cross-sectional observations (Figure 2b, bottom), and the thicknesses of the MXene layers with respect to the concentration were examined (Figure S9). As the concentration decreased, the thickness of the MXene layer decreased from 20 nm to 8 nm. To analyze the reliability of the electrical performances of the large-area MXene TFH, we checked nine equal-sized zones labeled 1 to 9, as shown in Figure 2c. The film exhibited an almost uniform temperature distribution, with a slight difference of approximately 2 °C between zones 5 and 1 (46.7 and 44.4 °C, respectively), despite the heat loss along the Ag contact electrodes. Additionally, the large-area MXene TFH had a nearly uniform distribution of sheet resistance values over the entire area. Figure 2d shows that the values were approximately 370 Ω/sq, with an average error range of ±15 Ω/sq.

Considering the advantage of the chemical compatibility of the MXene flakes, we developed an MXene heater incorporated with the PET substrate *via* a procedure identical to that for the glass substrate. As predicted, the MXene-coated PET film heater exhibited high bendability and flexibility, as well as optical transparency, owing to the typical structural advantages of 2D MXene flakes.[43] In Figure 2e and f, the bendable MXene TFH wrapped around the surface of the vial (r = 20 mm) exhibited uniform temperature distributions at 10 V, similar to those for the rigid TFH. We quantitatively investigated the heater performance of the flexible MXene TFH by evaluating the

DOI: 10.1021/acsnano.9b01602
ACS Nano XXXX, XXX, XXX−XXX

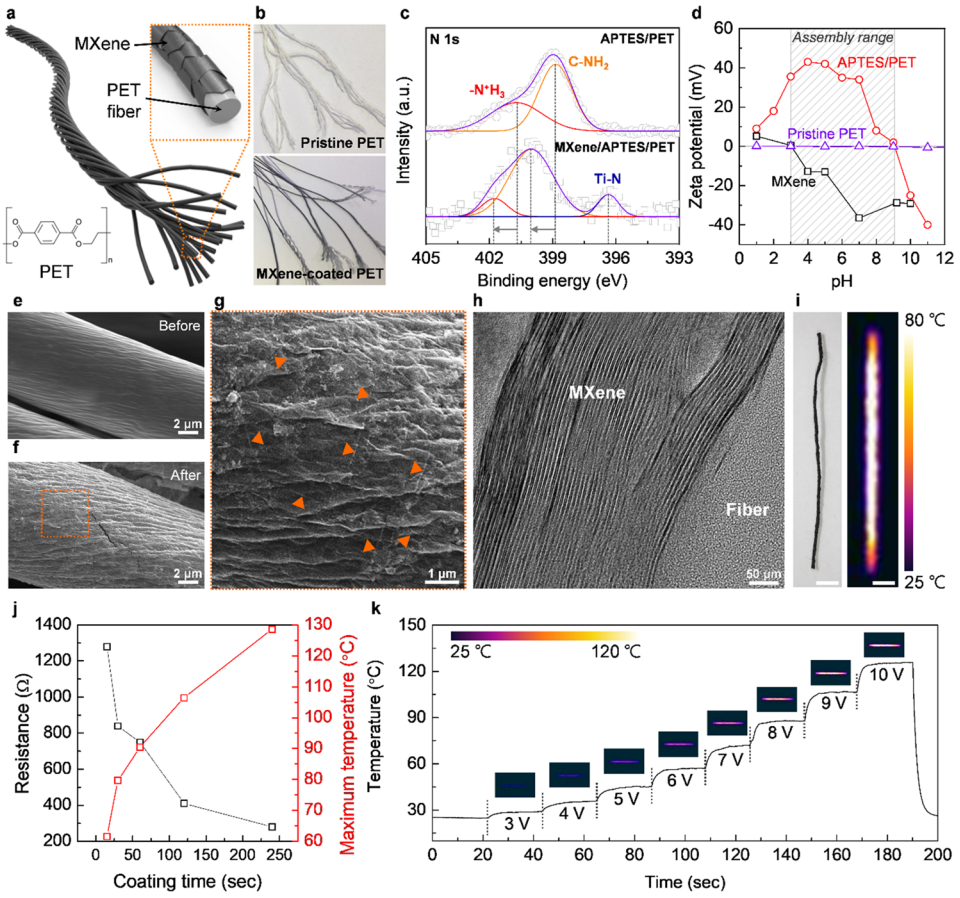


Figure 3. (a) Schematic of a PET fiber coated with MXene flakes. (b) Photographs of a pristine PET thread (top) and an MXene-coated PET TH (bottom). (c) N 1s XPS spectra of APTES-treated PET and MXene/APTES-treated PET. (d) Zeta-potential values of pristine PET, APTES-treated PET, and MXene, as a function of the pH. SEM images of (e) a pristine PET fiber and (f) the MXene TH. (g) Magnified image of the MXene TH. The orange triangles indicate the boundaries of the conformally coated layered MXene structures. (h) Cross-sectional TEM image of the MXene TH. (i) Photograph (left) and IR image (right) of the MXene TH operated at 10 V. All the scale bars represent 1 cm. (j) Resistance and maximum-temperature plots for the MXene TH with respect to the dip-coating time. (k) Temperature plots for the MXene TH at voltages ranging from 3 to 10 V. The inset shows IR images for each step.

resistance change ratio and temperature values with respect to the bending angle. Figure 2g shows that no significant resistance changes of the flexible MXene TFH were observed up to a bending angle of 90°. At bending angles of above 120°, the resistance of the flexible MXene TFH increased slightly, while the temperature gradually decreased. Finally, the temperature of the flexible MXene TFH decreased from 52.6 °C to 41.1 °C at a bending angle of 180°, but notably, the heater functioned even when folded perfectly in half. Interestingly, the flexible MXene TFH maintained its heating performance even after 100 bending cycles at an angle of 60°, without any delamination of MXene flakes (Figure S10).

MXene Thread Heater. The adaptability of the one-step coating process of MXene-dispersed water was investigated using various polymer threads, which are promising for next-generation wearable heaters. First, we used commercially available PET threads composed of hundreds of fibers. Before being coated with the MXene flakes, the PET threads were treated with O_2 plasma, followed by coating with (3-aminopropyl)triethoxysilane (APTES) to introduce amine groups for enhanced interaction (Figure S11). These processes enabled the PET threads to strongly electrostatically interact with terminal groups (T_x, T_x = −OH, −O, −F) of the MXene flakes without any apparent damage (Figure S12).[44] The APTES-functionalized PET threads were directly dipped in a water solution containing MXene flakes. Finally, the PET threads turned from white to black owing to the successful electrostatic interaction between the negatively charged $Ti_3C_2T_x$ and the positively charged APTES-functionalized PET, as schematically shown in Figure 3a. This interaction induced the self-assembly of MXene flakes on the individual PET fibers, resulting in closely packed surfaces of PET, whereas nontreated PET fibers inhibited the interaction of MXene flakes through repulsion force (Figure S13).

The photographs of the loosened thread in Figure 3b indicate that the MXene layer was perfectly fabricated on the surface of each fiber constituting the thread. In the N 1s XPS spectrum (Figure 3c and Figure S14), the APTES-treated PET exhibited the representative two peaks of C-NH_2 and protonated -N^+H_3 binding characteristics assigned at 398.86 and 400.71 eV, respectively, which means the surface of PET was successfully functionalized with amino groups.[45] After assembly between the APTES-treated PET and MXene flakes, both peaks were obviously shifted to higher energy. This was ascribed to the fact that a variety of negative moieties, such as −OH and −F, existing on the surface of MXene flakes enabled interaction with amine-functionalized PET, having positive surface charges through electrostatic interaction (Figure S11).[44] Also, in the

DOI: 10.1021/acsnano.9b01602
ACS Nano XXXX, XXX, XXX−XXX

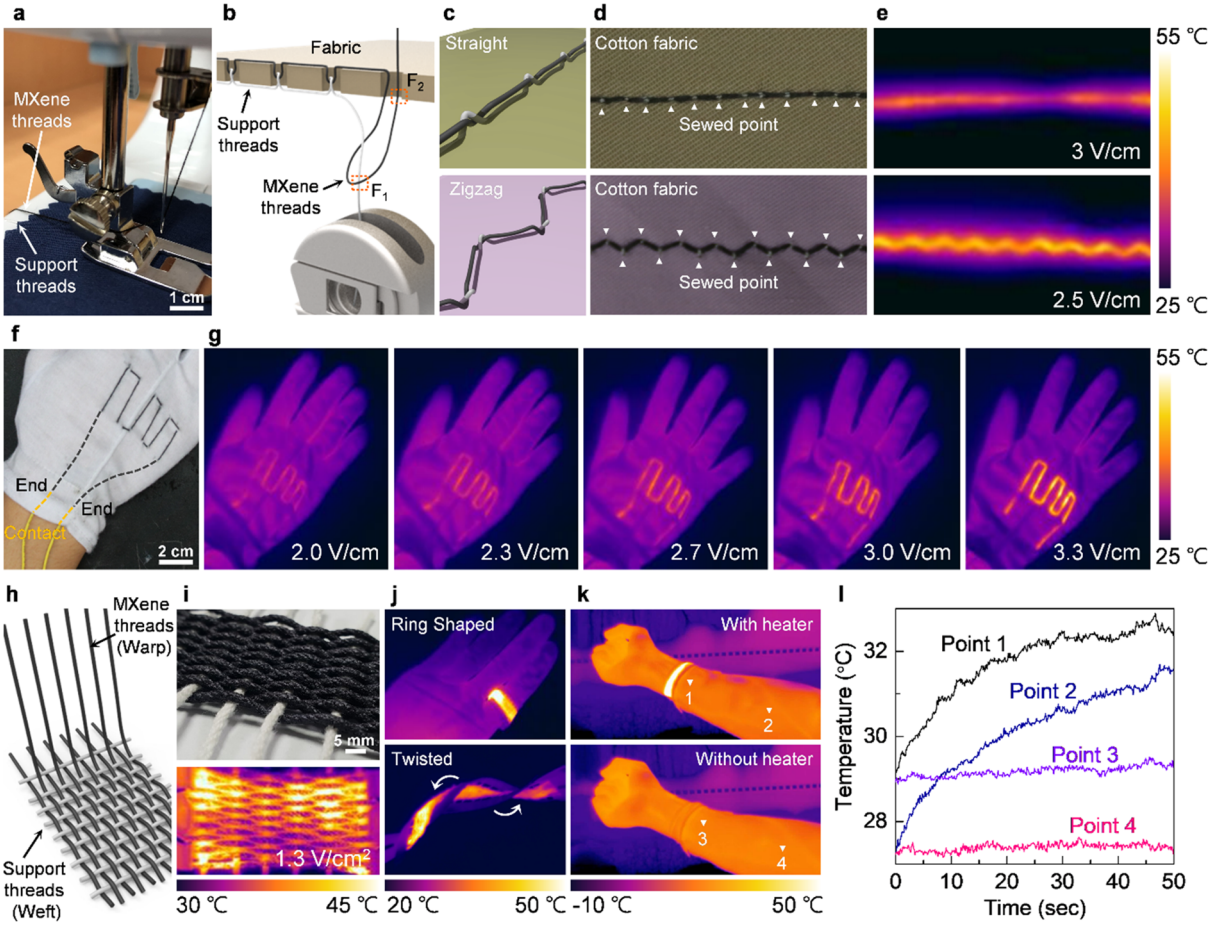


Figure 4. (a, b) Photograph and schematic illustration of the sewing-machine processing of fabricating an MXene heater containing MXene threads and support threads. The MXene threads and support threads were twisted by the rotating the bobbin and hook and then sewed onto the fabric. The friction forces (F_1 and F_2) are marked as dotted rectangles. (c) Schematics of the patterns with straight (top) and zigzag (bottom) shapes. (d) Photographs of the MXene TH sewed onto the cotton fabric with straight (top) and zigzag (bottom) patterns, where the sewed points are indicated by white triangles. (e) IR images of the MXene TH with straight (top) and zigzag (bottom) patterns at the applied voltage. (f) Photograph of a cotton glove sewed with the MXene TH. (g) IR images of the cotton glove sewed with the MXene TH at the applied voltages. (h) Schematic illustration of the weaving process, in which MXene THs and support threads were used as the warp and weft, respectively. (i) Photograph (top) and IR image (bottom) of the weaved MXene FH under 1.3 V/cm^2. (j) IR images of the MXene FH under different types of mechanical deformation: bending (top) and twisting (bottom). (k) IR images of the MXene FH worn on the arm with (top) and without (bottom) heating in an extreme freezing environment. (l) Temperature plot for each point on the arm (labeled 1 to 4 in (k)) with and without the MXene FH.

N 1s spectrum in MXene-coated PET, it was assumed that the existence of a newly appearing peak at 396.36 eV might arise from the interaction between an amino end group on the PET and the unstable titanium edge on the MXene, which is helpful for forming a solid layer composed of MXene flakes of the PET substrate *via* spontaneous interaction.[46]

Figure 3d shows the zeta-potential values of APTES-treated PET and the MXene/water dispersion, along with the pH. Although the PET initially had a slightly negative zeta-potential value without any surface treatment, the APTES-treated PET had a positive value below a pH of 8.6. Because the MXene/water dispersion had negative zeta-potential values at pH > 3.5, it was possible to produce a uniform assembly of MXene flakes and APTES-treated PET in the assembly range. The core−shell (thread−MXene) hybrid assembly occurred in the broad pH range of 3.5−8.6. Below a pH of 3.5 (acidic condition), the MXene flakes were easily peeled from the PET fibers, whereas above a pH of 8.6 (basic condition), individual PET fibers were strongly agglomerated with MXene flakes (Figure S15). Thus, the neutral pH condition was suitable for optimal assembly between the fibers and MXene flakes over the entire threads. Moreover, we investigated the morphological characteristics of a MXene-coated PET fiber with a diameter of 5 μm *via* electron microscopic analyses. Electrostatically assembled MXene flakes were uniformly coated even on the surface of a single PET fiber with a high topological roughness. Figure 3e and f show that the pristine PET fiber with a slightly wrinkled surface was coated with MXene flakes, and the coated surface had a similar topology to the pristine one. As observed under high magnification, the MXene flakes uniformly cover the intrinsic wrinkles of the PET fibers with evident edge lines, which are marked with orange triangles in Figure 3g. Figure 3h shows a cross-sectional image of the coated MXene flakes on the fiber surface, indicating that they perfectly maintained their initial layered structure. The MXene-coated PET TH with a length of 10 cm had a great heating performance at 10 V, as shown in Figure 3i, because the threads were composed of hundreds of MXene-coated single heaters. Our approach can be expanded to other commercially available polymer fibers, such as cellulose and nylon, owing to the

DOI: 10.1021/acsnano.9b01602
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advantageous compatibility of our pretreatment process for the enhanced interaction with MXene flakes (Figure S16).

To further analyze the heater performances of the MXene TH, we precisely controlled the dipping time of threads into MXene-dispersed water, where the concentration of MXene was fixed as 6 mg/mL. With the increase of the dip-coating time, the resistance values of the MXene-coated threads decreased by approximately 300 Ω, and the steady-state T_{max} values increased by 128.6 °C, as shown in Figure 3j (Figure S17). This is because the MXene flakes formed a coating layer with a controlled thickness on individual fibers of the threads. Figure 3k shows the stepwise temperature profile of our MXene TH with the voltage changing from 3 V to 10 V. The T_{max} values of the heater increased as the voltage increased, and the temperature rapidly became saturated (within approximately 10 s) in each voltage step.

Shape-Adaptable MXene THs. As shown in Figure 4a, the designed MXene THs with facile processability, high conversion performance of electrical to heat energy, and mechanical reliability were applied to cotton substrates using a conventional sewing machine. The prerolled pristine PET (support) threads were set free along the bobbin, and the MXene-coated PET threads released from the pinhole were hooked inside the sewing machine, as shown in Figure 4b. During this procedure, the MXene-coated PET threads were subjected to severe external stimuli, such as bending and friction. When the MXene-coated PET thread was sewed, it wrapped around the support thread with extreme bending stability (bending radius of about 0.5 mm). There were two frictional forces, which are denoted as F_1 and F_2. F_1 is the friction between the MXene threads and the support threads, and F_2 is the friction between the fabric and the MXene threads. There was no black contamination around the sewed points, indicating that the delamination of MXene from the threads rarely occurred (Supporting Video S1). Thus, the defined shapes were easily formed through the crossed node of the MXene THs and the support threads. Figure 4c and d show the successfully formed straight and zigzag shapes in the 2D space. Although the sewed TH was twisted with the rough-hewn insulator (supporter), the heating performance was observed along their formed shapes owing to the high durability of our MXene THs, as shown in Figure 4e.

More practically, our approach is suitable for carving our heater on cotton gloves for direct thermotherapy *via* local heating, as shown in Figure 4f and g. As the electric field increased from 2.0 V/cm to 3.3 V/cm, the temperature gradually increased, and T_{max} reached 53.5 °C. Finally, to evaluate the fabric-adaptable heater, we fabricated directly knitted MXene THs using a conventional knitting machine, where the MXene THs and the pristine threads acted as the warp and weft, respectively, as shown in Figure 4h. Figure 4i shows the MXene fabric heater (FH) composed of the arrays of MXene THs mechanically supported with the pristine threads as the weft. Because our FH has extraordinary flexibility owing to the intrinsic nature of cotton threads, it can provide good heating performance under different geometries (ring-shaped in the top image) and extreme deformation (double-twisted in the bottom image), as shown in Figure 4j. Finally, we examined the possibility of efficient thermotherapy by directly patching our heater on a human wrist. Our MXene FH was easily attached around the wrist owing to its high mechanical conformability, as shown in Figure 4k. The temperature change of the arm was measured in a harsh environment below 0 °C in order to observe the recovery of the body temperature during the heater operation. The temperatures at four points labeled 1, 2, 3, and 4 were profiled using an IR camera, as shown in Figure 4k and l. The MXene FH efficiently increased the body temperature by spreading the heat energy along the arm without damage to the skin.

CONCLUSIONS

We demonstrated that the $Ti_3C_2T_x$ MXene is highly efficient as an electric heater, with an intrinsically high electrical conductivity. Its 2D structure allows the facile formation of a thin coating layer on either films or threads treated *via* the chemical modification of amino-silanization, which increases the electrostatic interaction, regardless of the types of materials. The MXene-based heaters designed on a variety of substrates exhibited scalability, flexibility, processability, and expandability, as well as good performance and mechanical reliability. PET polymer threads composed of individual fibers coated with durable MXene flakes can be directly used to fabricate shape-adaptable and patchable heaters capable of operating even under extreme mechanical deformation induced by conventional sewing or knitting machines. Our results clearly suggest that our simple and robust heaters based on MXene flakes can be employed for various wearable/flexible applications, including healthcare thermotherapy.

EXPERIMENTAL SECTION

Preparation of MXene Thin Film. A Ti_3C_2 MXene solution was synthesized *via* etching of Ti_3AlC_2 powder (Carbon, Ukraine) according to the established minimally intensive layer delamination (MILD) method.[47] A 3 g amount of Ti_3AlC_2 MAX phase powder was reacted at 35 °C for 24 h with a mild etching solution, which was prepared by adding 4.8 g of LiF in 60 mL of 9 M HCl solution in a 250 mL polypropylene plastic bottle. The acidic product dispersions were washed copiously with deionized water (DI H_2O) *via* centrifugation at 3500 rpm for 5 min per cycle until a stable dark green supernatant solution of $Ti_3C_2T_x$ flakes reached a pH of ≥5. The final delaminated MXene aqueous solution was prepared by adding DI water to the clay-like sediment and then manually shaking it. MXene thin films were fabricated on glass and PET substrates. First, the substrates were cleaned with acetone and deionized water for 5 min each. Then, O_2 plasma was applied to them at a power of 60 W for 5 min, followed by immersion in 2% v/v APTES (99%, Sigma-Aldrich, USA) and ethanol for 30 min to generate NH_2 groups. After the surface treatment, the aqueous dispersion of MXene with concentrations of 3, 4, 5, 6, and 7 mg/mL was spin-cast onto the substrates at 1000 rpm for 10 s and then 2000 rpm for 5 s, respectively. Finally, the resulting film was annealed at 200 °C in a N_2-filled box for 2 h.

MXene Coating on Polymer Threads. Commercially available PET, cellulose, and nylon threads were used as substrate fibers. Before the coating, O_2 plasma was applied at 60 W for 5 min, followed by immersion of the threads in 2% v/v APTES (99%, Sigma−Aldrich, USA) and ethanol for 30 min to generate NH_2 groups. The APTES-functionalized threads were immersed in the aqueous dispersion of MXene with a concentration of 6 mg/mL for dipping times of 15, 30, 60, 120, and 240 s. Finally, the resulting threads were annealed at 200 °C in a N_2-filled box for 2 h.

Sewing and Knitting of MXene-Coated Polymer Threads. A sewing machine (HSSM-3000, Hons Korea, Korea) was used to sew the MXene-coated PET threads with the given shapes on a cotton fabric substrate (5 × 8 cm^2). The sewing was performed by hooking the two types of threads composed of pristine PET threads (support) and the MXene-coated PET threads with straight and zigzag shapes. For practical application, the threads were also sewed on a cotton glove in the shape of a trident. A knitting machine (Deluxe Amylin, Bandai, Japan) was used to fabricate the fabric-type MXene heater. Similar to the sewing procedure, the pristine cellulose threads (supports) and the

DOI: 10.1021/acsnano.9b01602
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MXene-coated cellulose threads acted as the weft and warp, respectively, for knitting in a perpendicular direction.

Electrical, Thermal, and Electrothermal Properties of MXene-Based Heaters. The sheet resistance of the MXene heaters was measured using a four-point probe system (AIT, CMT-100MP, USA). For the heating test, both sides of the heaters were coated with Ag paste (ELCOAT P-100, Japan), and an electric potential was applied across the two electrodes using a waveform generator (Agilent 33220A, Agilent, USA). For a better understanding, the applied voltage values in sewed and knitted heaters were normalized with the length and area of the medium, respectively. The thermal properties were monitored, and images with pseudocolors were obtained using an IR camera (FLIR A310, FLIR, Sweden). For the deicing test, the MXene TFH was immersed in liquid nitrogen for 5 min to reach an isothermal state, followed by an inducing potential of 12 V. For the thermotherapy test, an arm equipped with an MXene FH was placed in a box set to −5 °C to reach an isothermal state, and the temperature changes of the two positions away from MXene FH were recorded for 50 s.

General Characterization. Surface morphology of the MXene layer was analyzed using field-emission SEM (Inspect F50, FEI, USA). A cross-sectional view of the layer was obtained using TEM (Titan 80-300, FEI, USA) equipped with a focused ion beam system (Nova 600 NanoLab, FEI, USA). The optical transmittance was measured using an ultraviolet−visible spectrometer (V-650, JASCO Corporation). The surface charge was determined using an electrophoretic measurement apparatus (ELSZ-1000, Otsuka Electronics Co., Ltd., Japan) at the given pH values. The zeta potentials of the pristine PET, APTES-treated PET, and MXene-coated PET film samples were measured using a solid sample cell configured with a box-like quartz cell. The potential values were achieved by analyzing the apparent electrophoretic mobility on several positions of the samples based on the Mori−Okamoto equation. The thin-film HR-XRD patterns were recorded using a Rigaku SmartLab equipment. Chemical compositions of the MXene TFHs were identified using the XPS system (model K-Alpha, Thermo Scientific, USA). Raman spectra was collected using ND:Yag 532 nm excitation laser (LabRam Aramis, Horriba Jovin Yvon).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsnano.9b01602.

rGO properties recently reported in the literature, spectral transmittance, maximum temperature as a function of V and $1/R$ of MXene TFH, temperature plots during defrosting at voltage on and off, TEM images, durability properties during bending cycles, scheme of plasma treatment and dip coating, SEM image comparison of with and without treatment, EDS spectrum and elemental mapping of MXene TH, SEM images of MXene TH, performance of cellulose and nylon-6 THs, maximum temperature with respect to the input voltage of MXene TH (PDF)

Movie (AVI)

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Author Contributions

T.H.P. and S.Y. conducted most of the experiments including fabrication of devices, analysis of data, and preparation of the manuscript. M.K. contributed to the application of devices and preparation of the manuscript. H.K. contributed to synthesis of MXene flakes and analysis of the data. E.H.K. contributed to development of the MXene thin film. J.-E.P. conducted the experiment with the sewing and weaving machine. B.O. measured zeta potentials and observed SEM images. B.K. and S.H.N. set up the equipment and gave advice on the measurement. C.H.P. conducted XRD and Raman measurements for the stability of MXene TFH before and after heating/cooling cycles. All authors discussed the results and contributed to the paper.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This project was supported by a grant from the National Research Foundation of Korea (NRF) funded by the Korean government (MEST) (No. 2017R1A2A1A05001160) and the Creative Materials Discovery Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT (2018M3D1A1058536). This work was partially supported by a grant from the Basic Science Research Program (2017R1A2B3006469) through the National Research Foundation of Korea funded by the Ministry of Science, ICT, and Future Planning. This research was also supported by the Korea Electrotechnology Research Institute (KERI) Primary research program (No. 19-12-N0101-17) and Korea Institute of Science and Technology Primary research program and KU-KIST program.

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